The JUNO Reactor Neutrino Experiment

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I Abstract

The Jiangmen Underground Neutrino Observatory (JUNO) is a 20kton liquid scintillator detector located 700m underground. JUNO's outstanding energy resolution and large fiducial volume offer the opportunity to address pressing questions in neutrino Physics. It is widely anticipated that the neutrino Mass Hierarchy (MH) will be determined this decade, at a 3-4σ significance level after 6 years of data taking at JUNO. Obtaining the spectrum of antineutrinos produced by nearby nuclear power plants to such a high precision will lead to improved determination of the neutrino oscillation parameters $\sin^2\theta_{12}, \Delta m^2_{21}$ and $|\Delta m^2_{ee}|$. The focus of this review is the determination of the MH at JUNO, acknowledging the implications this can have in other open questions such as the Majorana or Dirac nature of neutrinos and discusses how this can be done, concluding that the MH determination could benefit from other measurements including those from JUNO for a more robust measurement. JUNO has great potential for further advancing our quest in understanding the fundamental properties of neutrinos, one of the building blocks of the Universe.

1 Introduction

The JUNO experiment was proposed in 2008 with the primary goal of determining the neutrino mass hierarchy (MH). This would be allowed for by detecting reactor antineutrinos from nuclear power plants. Determination of the MH requires equal baselines from the detector to the reactor cores so that the oscillation dephasing effect can be seen. This method was first proposed in 2002 by Petcov [1]. The non-zero θ_{13} angle has been discovered [2] in both reactor and accelerator neutrino experiments, making the aforementioned method feasible. JUNO will be the first experiment to measure solar and atmospheric oscillation effects simultaneously and will be the largest liquid scintillator (LS) detector ever built, containing 20 kilotonnes of liquid scintillator, monitored by ∼17600 20" high quantum efficiency PMTs and ∼25600 3" PMTs providing a total coverage of ∼78%.

The extraordinary performance of the detector and the large statistics obtained from the reactor antineutrino mean that the $\sin^2\theta_{12}$, Δm^2_{21} and $|\Delta m^2_{32}|$ oscillation parameters will be calculated to better than 0.6% precision in six years of data taking [3]. JUNO is located ∼50km from the Yangjiang NPP and Taishan NPP. The JUNO detector has a central detector, a muon tracker and a water Cherenkov detector. PMTs will detect the Cherenkov light from cosmic muons. The central liquid scintillator (LS) detector is designed to have an energy resolution of 3 % / $\sqrt{(E(MeV)}$. Obtaining such an energy resolution is a very difficult task.

Thanks to a great effort from a number of different experiments, we can say that the three-flavour neutrino oscillation has three non-zero mixing angles [4] in the MNSP matrix [5] and two independent mass-squared differences $|\Delta m^2_{31}| = |m^2_3 - m^2_1|$ and $|\Delta m^2_{32}| = |m^2_3 - m^2_2|$ have been measured. The neutrino MH, the octant of the mixing angle θ_{23} and the leptonic CP-violating phase δ in the MNSP matrix remain unknown, and will be addressed by the next generation of long-baseline experiments such as DUNE and Hyper-K. These are the main objectives of proposed neutrino oscillation experiments.

2 Mass Ordering

JUNO measurements do not rely on the matter effect in neutrino oscillations or δ_{cp}, but instead on the oscillation interference between Δm^2_{31} and Δm^2_{32} with $\Delta m^2_{ij} = m^2_i - m^2_j$. It can be seen that the squared mass differences thus depends on the mass ordering.

The results of the SNO experiment determined that $m_1 < m_2$, but the complete ordering of the mass eigenstates ν_1, ν_2, ν_3 remains unknown. Both normal hierarchy (NH); $m_1 < m_2 < m_3$ and inverse hierarchy (IH); $m_3 < m_1 < m_2$ are compatible with the current data [6] [7] and thus we are presented with one of the most pressing Physics problems of our time.

The determination of the MH has many implications on our understanding of neutrino physics, astronomy and cosmology. The MH will help guide the $0\nu\beta\beta$ experiments, which could reveal the Dirac or Majorana nature of neutrinos. For an IH conclusion there is a greater probability for the observation of $0\nu\beta\beta$ and the Majorana nature of neutrinos. The determination of MH will also be needed to determine the lepton CP-violating phase and is key to our understanding of neutrino mass generation. Additionally the NH could be attributed to the existence of GUTs due to the similarities of quarks and leptons. An IH predicts a degenerate spectrum between the first and second eigenstates of mass. This could be explained by models which predict discrete or U(1) flavour symmetries. Determination of MH is essential to understand the origin of neutrino mass and the mixing phenomenon.

3 JUNO Measurements

To obtain a resolution of 3% / $\sqrt{E(MeV)}$, JUNO must detect more than 1200 photoelectrons per deposited MeV. To achieve this, firstly the photon light yield and attenuation length of the scintillator are increased. An improved photocathode coverage of more than 75%, using 18000 50cm PMTs, can be reached and a high detection quantum efficiency (30%) of the used multipliers has been achieved. For context one of the Super-K PMTs would reach around $\sim$ 15% efficiency. The adopted calibration strategy uses a novel dual calorimetry technique, outlined in [8], to assist in obtaining the required energy resolution. There are many design considerations for the success of such an experiment, such as radioactivity effects. In this case the recommended strategy for JUNO is detailed in [9], for example selecting the construction materials according to their intrinsic low concentration of natural radioactive species. These are examples of considerations to take into account to minimise impact on neutrino mass ordering determination.

Why is such a high resolution required? The oscillation interference manifests in Δm_{31} and Δm_{32} as $\frac{\Delta m^2_{21}}{|\Delta m^2_{32}|}\sim$3%. The reactor spectrum must have a precision better than 3% or the effects will be washed out by the energy resolution of the detector.

JUNO is a medium baseline experiment, meaning that the oscillation of the mass-squared difference is contained in the energy spectrum as the multiple cycles. This is done by considering the electron antineutrino survival probability, as shown in [10]. This equation considers the vacuum case, but in analysis of real data the matter effects should be included.

$$P_{\bar{\nu}_e\to\bar{\nu}_e} = 1 - \frac{1}{2}\sin^2 2\theta_{13}[1 - \sqrt{1-\sin^2 2\theta_{12}\sin^2\Delta_{21}}\cos(2|\Delta_{ee}| \pm \phi)] - \cos^4\theta_{13}\sin^2 2\theta_{12}\sin^2\Delta_{21} \tag{1}$$

The $\pm$ sign in the last term of equation 1 is decided by the MH (+ for NH and - for IH).The spectral distortion holds the MH information, conveyed by the energy and baseline dependence of the extra effective mass-squared difference Δm^2_ϕ [11].

$$\Delta m^2_\phi = \frac{4E\phi}{L} \tag{2}$$

At this baseline, Δm^2_ϕ is greater at lower energies. For the case of NH, the effective mass squared difference in Equation 1 at low energies will be greater than at high energies and the converse case for IH, due to the changing sign. Figure 1 shows the difference in mass ordering and therefore how this can be determined with JUNO. It has been shown that this difference is more visible in the frequency domain via a Fourier transform of the spectrum of antineutrinos [12], [13].

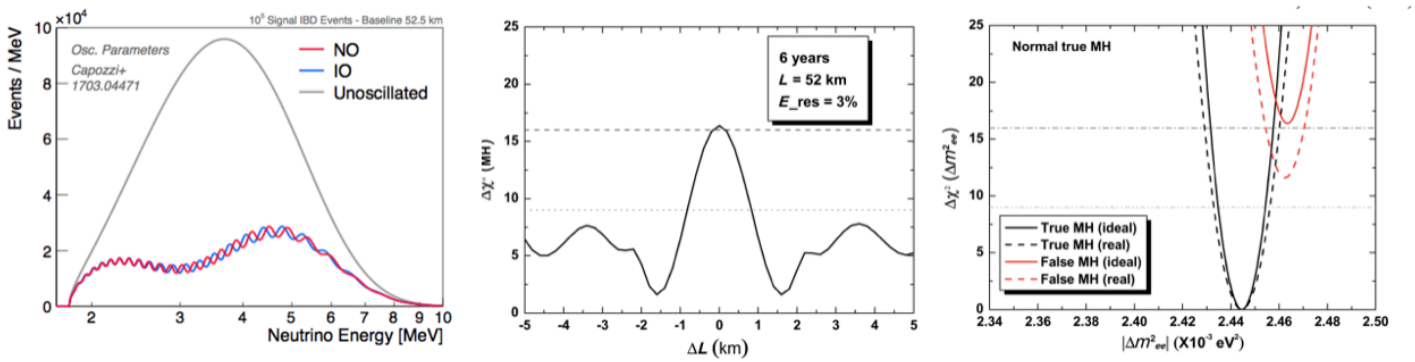

Figure 1: *The relative shape difference of the reactor antineutrino flux for different MHs (left). Variation of MH with baseline difference of two reactors (middle) and the effect on MH sensitivity for ideal and actual reactor core distribution. True = NH and False =IH Different oscillation patterns from NH and IH (right). [12]*

In addition to the interference of $|\Delta m^2_{31}|$ and $|\Delta m^2_{32}|$, the measurement of $|\Delta m^2_{ee}|$ can provide further insight into the MH alongisde the $|\Delta m^2_{\mu\mu}|$ oiginating from the muon neutrino dissapearance. By obtaining both these measurements, JUNO has a clear path to answering this question.

As has been shown in [14] and [15], the long baseline, high statistics and energy resolution at JUNO yield a sensitivity which is an order of magnitude better than previous reactor experiments such as Daya Bay [16], KamLAND [17] and RENO [18].

4 Systematics

To assess if JUNO can viably measure the MH, it's crucial to consider the effect of systematics involved. This could include accounting for reactor, detector, background and energy related uncertainties.

A dedicated calibration of the detector energy nonlinearity response is another critical factor to obtain reliable sensitivity of the MH determination. The uncertainty from the detector nonlinearity response can distort the antineutrino spectrum and is crucial for JUNO's success, since a precise energy spectrum of reactor antineutrinos is required to resolve the MH. The issue was first pointed out in [19] and has been accounted for in [10], concluding limited impact on mass ordering. Other contributing factors such as matter effects, background, energy resolution and scale have been investigated. Statistics are very important to consider. The test statistics to determine MH are shown to be normally distributed [20].

It's evident that the nature of the experiment makes it a difficult challenge to establish MH. With 8 years of data, the difference in the number of events for NH and IH is merely tens of events per energy bin which is smaller than the statistical uncertainty in each bin. Pseudo experiments in [21] show that after 8 years, in only 31% of the trials can JUNO determine the neutrino mass ordering at the level of 3σ or better. This paper also finds that there is an 8% probability that the wrong mass ordering is found. This paper uses the $\bar{\chi}^2_{min}$ with normal ordering assumed, then for IH the larger this value, the larger the confidence level one can discriminate between NH and IH using JUNO. Also highlighted is the influence of the true value of Δm^2_{21}. A value either smaller or larger than the currently accepted could shift $\bar{\chi}^2_{min}$ to either 7.1 or 10.2 from 8.5 which is immensely significant. This paper points out that an improvement on the energy resolution from 3% to 2.9%, a challenging feat to achieve, would represent an increase of $\bar{\chi}^2_{min}$ of IH from 8.5 to 9.7. Clearly it is important to consider the impact of these factors.

Site selection is a determining factor which was chosen carefully for JUNO's MH determination as is explored in [10]. A medium baseline experiment is ideal for MH determination. A short or long baseline experiment makes it impossible to distinguish between NH and IH because two degenerate solutions with identical mass squared differences but different MH can manifest as identical neutrino energy spectrums. The oscillation maximum of θ_{12} is the most sensitive position for the determination of MH. This location is a source of sensitivity loss in JUNO. If the baseline is off by 1km, we can lose the sensitivity completely as seen in Figure 1. Therefore it's key find perfect location with baseline off no more than 500m, a challenge due to the clustered nature of the reactor cores.

5 Conclusion

Determining the neutrino MH is an important task for neutrino Physicists to persue, as discussed in this review. The true MH provides crucial information for future endeavours in Physics, for example $0\nu\beta\beta$ decays and building models of neutrino masses and flavour mixing. The sensitivity of the MH measurement is strongly dependent on the energy resolution, baseline differences and energy response functions as seen in Section 4. Along with the measurements offered by different experiments such as Super-K, a conclusive determination with a sensitivity of 3σ is expected to be obtained. Such a breakthrough in neutrino physics deserves the effort of multiple experiments with different techniques, to provide a robust measurement and confidence in the MH determination. Various factors will play a role in the accuracy of the measurement. We remain unable to say, with attempts to prefer one order seemingly unsignificant [7], and so the question remains open.

It was discussed in Section 4 how [10] has shown that precision measurements are crucial for MH determination. Obtaining such a high energy resolution is challenging, but key to success. Also discussed is the other unknowns mentioned in Section 1, for example the ambiguity of the unknown CP-violating phase and the consequent effect on the MH sensitivity with respect to change in $\Delta m^2_{\mu\mu}$ error. NoνA and T2K will be able to obtain high precision measurements of this error, useful for JUNO.

The ability of JUNO to determine the MH at $\geq 3\sigma$ to be 31% is assessed in [21], demonstrating the importance of statistics and showing that this probability is particularly sensitive to the true values of the oscillation parameters, especially Δm^2_{21}. However, combining JUNO's measurement of $|\Delta m^2_{ee}|$ with other experiments in a global fit is shown to provide an earlier determination of the MH, also concluded in [10]. If JUNO's precise measurement of $|\Delta m^2_{ee}|$ is combined with measurements from accelerator and atmospheric neutrino experiments (e.g T2K, NOvA, Super-K), then the MH is likely to be determined at $\geq 3\sigma$ much sooner than without, possibly just 2 years after JUNO starts data taking. This demonstrates the importance of a variety of experiments involved in the common goal to uncover knowledge of unknown neutrino Physics.

Additionally, JUNO will register ~ 5000 events from inverse beta decay events $\bar{\nu}_e + p \rightarrow n + e^+$ (comparable to Super-K) and will also register 300 events from neutrino-electron scattering, making it the best supernova ν_e detector so far [22]. This measurement could further robustness as flavour conversion of supernova neutrinos can give insight into the nature of the MH. Neutrinos traversing the supernova will encounter a range of matter densities. This allows for MSW conversions [23] driven by the Δ^2_{31} mass difference and the θ_{13} mixing angle, and consequently by the Δ^2_{21} and the θ_{12}. This is discussed further in [24].

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